

Human-Computer Interaction

**Intro to
CS 770**

Professor Bilge Mutlu

Today's Agenda

- Who we are
- The three strands, decoupled by day
- *Fundamentals are the medium* – the AI framing
- Syllabus, grading, crit model, AI-use policy
- Seeding the project + topic cards

Your Instructor

Bilge Mutlu

*Professor of Computer Sciences,
Psychology, & Industrial Engineering*

PhD, 2009, Carnegie Mellon University

bilge@cs.wisc.edu · bilgemutlu.com

Office hours: Fridays 11 am-12 pm
(location TBD)



Your TA

Hailey Johnson

*PhD candidate, Computer Sciences –
People and Robots Lab*

hljohnson22@wisc.edu ·
haileyjohnson.github.io

Office hours: Tuesdays 1–2 pm (*location
TBD*)



How about you?
Give us your name, program,
year.

Three strands, decoupled by day

Monday

Seminar – one enduring HCI topic per week

Friday

Methods – M1-M12, the research toolkit

Across both

Project – a term-long team study

Why decouple the strands?

The gap is the point.

- Seminar gives you the **questions** worth asking
- Methods give you the **instruments** to answer them
- The project is where they **collide** – you practice both on one real study

Watch this

A researcher delegates a literature search, a schedule, and a collaboration – to an agent.

As you watch, track one thing: *what does the **person** actually do?*

► [play \(YouTube\)](#) · [backup: Internet Archive](#)



1987.

Why doesn't it exist yet?

The durable questions

- **Delegation** – what do you hand over?
- **Common ground** – what does it know that you know?
- **Trust & reliance** – when

The barriers, 39 years on

- **Privacy** – always listening; no wake word, ever
- **Social & situational** – voice fails in an open office
- **Trust** – would you let it

^{kn} Newendorp, A.K., Sanaei, M., Perron, A.J., Sabouni, H., Javadpour, N., Sells, M.J., Nelson, K.T., Dorneich, M.C., & Gilbert, S.B. (2024). Apple's Knowledge Navigator: Why Doesn't that Conversational Agent Exist Yet? *CHI '24*, Article 776, 14 pp. doi:10.1145/3613904.3642739

**Fundamentals are the *medium*,
not the message**

The AI framing

This is a **research** course that uses enduring HCI fundamentals as the instrument for reading the current AI moment.

Every seminar week runs one frame:

Enduring fundamental → how AI is changing it → depth (a claim to argue with)

Two rules that hold all term

- Each seminar closes with a **guardrails coda** – oversight, accountability, appropriate reliance.
 - Design & usability content enters only as **"lens in, craft out"** – a theoretical lens or a research method, **never** as practice craft.
- 👉 We are not here to *build* good interfaces. We are here to *study* interaction.

What is HCI research?

"...a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."

– ACM

770 lives in the second half: *the study of major phenomena.*

570 vs. 571 vs. 770

770

- Research methods
- Grads across campus
- Project-based
- No technical background

570

- Design methods
- Undergrads
- Project-based
- No technical background

571

- Design/building
- CS undergrads
- Assignment-based
- Needs CS-400 & JS

Learning Goals – *asking*

- Pose research questions, situate them in the literature, and identify gaps
- Meet seminal HCI research across topics
- Match a research approach to a question – qualitative, quantitative, hybrid

Learning Goals – *doing*

- Choose measures; design surveys & scales; assess reliability and validity
- Analyze qualitative and quantitative data
- Carry out an original project and **write it up as a paper**

Setting Expectations

- Read a lot – **1-3 papers + a book chapter** most weeks
- Budget **10-15 hrs/week** (a real 3-credit graduate load)
- A substantial **semester-long team project**
- Come ready to **discuss** – this is a high-interactivity room

Course Policies

Grading

Assessment	Points
Seminar participation	20
Methods hands-on activities	30
Project milestones	24
Final paper	8
Final presentation	8
Attendance & participation	10
<i>Total</i>	<i>100</i>

Attendance & participation (10 pts)

- **Attendance (5)** – seat check against a fixed pod chart; **2 free absences**, no questions asked
- **Pod role (3)** – four roles rotate in your pod each Monday:
Reporter · Synthesizer · Skeptic · Connector
- **Self-assessment (2)** – two half-page reflections (Oct 26, Dec 7)
making your case **with evidence**

The crit model

You'll critique each other's work at the **method-design** and **analysis** milestones – self-run, in your pod.

- The **first crit is modeled live** so you see how it should go
- Etiquette: **specific / actionable / kind** – critique the work, not the team
- Your pod is **three teams**, self-run on a timer I call from the front:
4 min present · **2 min** silent written feedback · **5 min** round-robin
· **2 min** respond & commit

AI-Use Policy

Use AI to do your best work – not to cut corners.

- You are **expected** to use it. No disclosure notes on coursework.
- What matters isn't which tool you touched – it's whether the **intellectual contribution** is yours.

Mind the jagged frontier

AI is excellent at some tasks and **confidently wrong** at neighboring ones – and the boundary is invisible, and it moves.

- **+30%** quality on tasks *inside* the frontier
- **–19 points** correctness on a task *outside* it

Learning where that edge falls for **your** work is part of what you're here to develop.

^{ai1} Dell'Acqua, F., McFowland, E., Mollick, E., Lifshitz, H., Kellogg, K.C., Rajendran, S., Kraye, L., Candelon, F., & Lakhani, K.R. (2026). Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of Artificial Intelligence on Knowledge Worker Productivity and Quality. *Organization Science*, 37(2), 403–423. doi:10.1287/orsc.2025.21838

The two hard lines

You own what you submit

Every claim, number, and citation is yours to stand behind.

Fabricated data, results, or citations are academic misconduct – regardless of what produced them.

Venue rules override mine

Journals and conferences set their own requirements, and they change. **Our final paper follows ACM's** – generative-AI use disclosed in a brief Acknowledgements note.

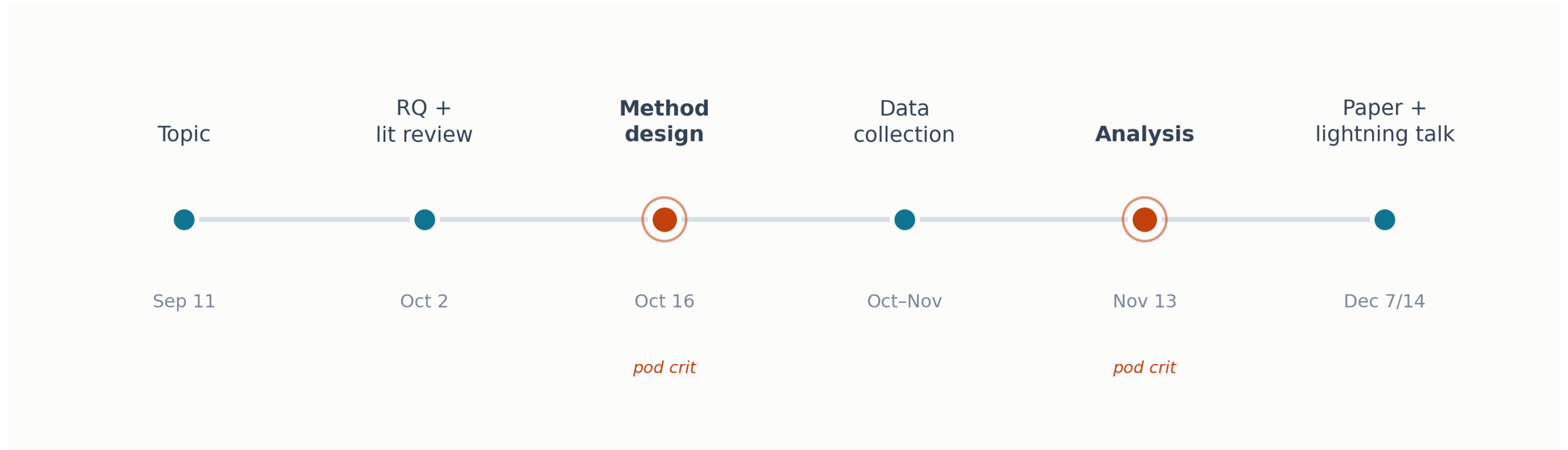
The Project

The project, end to end

A term-long, **3-person** research project – the place the seminar and methods strands meet.

- **Topic → RQ / lit review → method design → data → analysis → paper + lightning talk**
- Incrementally written up as a full **ACM CHI-format paper**
- Mixes qualitative *and* quantitative methods

Six milestones, two crits



Seeding the project (*today only*)

We're **priming**, not forming.

- Take home the **speculative-design topic cards** as a thinking aid
- Let ideas marinate over the week – no topics, no teams yet
- Everything real happens **next Friday (Sep 11)**: topic selection, team formation, standing-pod assignment

Assigned readings (*for Fri Sep 11 / M1*)

- **Lazar, Feng & Hochheiser (2017), Ch. 1** – Introduction to HCI research
- **Grudin (2012)** – *A Moving Target: The Evolution of HCI* (survey depth)
- **Wobbrock & Kientz (2016)** – *Research Contributions in HCI* – the seven types

What's Next?

- **Fri Sep 11 – M1: What is HCI research? + Project kickoff**
 - Read **Lazar et al. Ch. 1**
 - Bring your **topic cards**
 - Come ready to **pick a topic and form a team** – pods get assigned